

WIRELESS NETWORK VULNERABILITY ASSESSMENT REPORT

1.TJX Companies Wireless Network Breach, 2005–2007

The TJX Companies data breach, which occurred from 2005 to 2007, was one of the most significant payment-card security incidents of its time. Attackers gained unauthorized access to TJX's retail network through wireless vulnerabilities and subsequently obtained large quantities of payment-card information. The incident exposed weaknesses in wireless encryption, authentication, network segmentation, security monitoring, and protection of sensitive payment data.

1. Identified Vulnerabilities

- Obsolete WEP encryption and weak wireless protection
- Insufficient authentication and access control
- Inadequate separation of wireless users from payment systems
- Limited wireless monitoring and rogue-access-point detection
- Insufficient protection of sensitive payment-card data

2. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

3.Commands

```
nmcli device wifi list
sudo iw dev wlan0 scan
sudo airodump-ng wlan0
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.10
sudo nft list ruleset
```

4.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

2.Lowe’s Wireless Network Intrusion, 2003

1. Incident Overview

A compromised retail wireless connection reportedly provided a path toward central corporate systems, illustrating the danger of connecting store wireless networks too closely to sensitive enterprise and point-of-sale environments.

2. Identified Vulnerabilities

- Weak wireless authentication
- Insufficient wireless isolation
- Inadequate firewall boundaries between store and corporate systems
- Insufficient POS segmentation
- Limited intrusion detection and centralized access management

3. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

1. Use enterprise WPA2/WPA3 with 802.1X and unique identities rather than shared wireless credentials.
2. Place POS systems in isolated, tightly controlled network segments.
3. Use firewalls between wireless, store, POS, and corporate networks with deny-by-default rules.
4. Deploy IDS/IPS and centralized security monitoring for unusual lateral movement.
5. Apply centralized identity, privileged-access management, logging, and regular access reviews.

5. Overall Risk Rating

Critical

6.Commands

```
ip addr
iw dev
iwconfig
nmcli device wifi list
sudo iw dev wlan0 scan
sudo airmon-ng
sudo airmon-ng start wlan0
```

```
sudo airodump-ng wlan0mon  
ip route  
sudo nmap -sn 192.168.1.0/24  
sudo nmap -sn 192.168.10.0/24  
sudo nmap -sV 192.168.1.10  
sudo nft list ruleset  
sudo iptables -L -n -v  
sudo ss -tulpn  
sudo nmap --script vuln 192.168.1.10  
sudo wireshark
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

3.Firesheep and Public Wi-Fi Session Hijacking, 2010

1.Incident Overview

Firesheep demonstrated that users on the same open wireless network could capture unencrypted session cookies and impersonate authenticated users when applications failed to protect the entire session with HTTPS.

2.Identified Vulnerabilities

- Open wireless networks exposing traffic to nearby clients
- Session cookies transmitted without adequate encryption
- Incomplete HTTPS enforcement
- Missing or weak HSTS
- Insufficient client isolation and security awareness

3.Vulnerability Assessment

Area	Assessment	Risk
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Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4.Recommended Security Controls

6. Enforce HTTPS across every authenticated page and resource and redirect HTTP to HTTPS.
7. Use HSTS with an appropriate policy and secure cookie attributes such as Secure and HttpOnly; use SameSite where appropriate.
8. Enable wireless client isolation so connected users cannot directly communicate with each other.
9. Recommend or provide VPN protection for sensitive activity on untrusted networks.
10. Educate users to verify HTTPS/certificate warnings and avoid transmitting sensitive information over untrusted Wi-Fi.

5.Overall Risk Rating

High

6. Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

3.DarkHotel Attacks Through Hotel Wi-Fi Networks

1. Incident Overview

The DarkHotel campaign illustrated how compromised or untrusted hotel networks can be used to target selected guests with fraudulent software-update prompts and malware.

2. Identified Vulnerabilities

- Weak guest authentication and captive-portal trust
- Potential DNS or traffic manipulation
- Insufficient guest-network segmentation
- Third-party management and privileged access risks
- Insecure software-update delivery or inadequate signature validation
- Insufficient malware and network monitoring

3. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

11. Isolate guest Wi-Fi from hotel administration, POS, surveillance, and building-management networks.
12. Secure captive portals with HTTPS, strong certificates, minimal data collection, and protected administrative interfaces.
13. Use DNS security controls and monitor for unexpected DNS changes or malicious domains.
14. Require signed software updates and endpoint verification; never trust update prompts solely because they appear on a hotel network.
15. Restrict third-party access with MFA, least privilege, time-limited accounts, logging, and periodic review.
16. Deploy wireless monitoring, endpoint detection, and centralized incident response.

5.Overall Risk Rating

Critical

6.commands

```
ip addr
iw dev
iwconfig
```

```
nmcli device wifi list
sudo iw dev wlan0 scan
ip route
sudo nmap -sn 192.168.1.0/24
sudo nmap -p 80,443 192.168.1.1
sudo nmap -sV 192.168.1.1
sudo ss -tulpn
sudo nft list ruleset
sudo iptables -L -n -v
sudo tcpdump -i wlan0 -c 100
sudo tshark -i wlan0 -c 100
sudo wireshark
curl -I https://example.com
curl -I http://example.com
curl -I -L https://example.com
```


7. Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

4.Marriott Wi-Fi Deauthentication Incident, 2014

1. Incident Overview

The incident concerned use of equipment to transmit Wi-Fi deauthentication messages that disrupted guests' personal hotspots, demonstrating an availability risk associated with unprotected 802.11 management frames.

2. Identified Vulnerabilities

- Management frames vulnerable to spoofed deauthentication
- Wireless availability and interference risk
- Insufficient wireless intrusion detection/prevention
- Inadequate acceptable-use governance
- Potential regulatory and legal exposure

3. Vulnerability Assessment

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Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

17. Use Protected Management Frames (802.11w/PMF) where supported.
18. Prefer WPA3 configurations that require or strongly support PMF according to the deployment and client capabilities.
19. Deploy wireless IDS/IPS capable of detecting abnormal deauthentication and interference patterns.
20. Define acceptable-use policies that clearly prohibit deliberate disruption of guest communications.
21. Maintain documented authorization, testing, escalation, and incident-response procedures for wireless security tools.
22. Assess regulatory requirements and legal consequences before deploying controls that can affect third-party communications.

5. Overall Risk Rating

High

6.commands

```
ip addr
iw dev
iwconfig
```

```
nmcli device wifi list
sudo iw dev wlan0 scan
ip route
cat /etc/resolv.conf
resolvectl status
nslookup example.com
dig example.com
dig +short example.com
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.1
sudo nmap -sV 192.168.10.1
sudo nft list ruleset
sudo iptables -L -n -v
sudo ss -tulpn
curl -I http://example.com
curl -I https://example.com
curl -sI https://example.com
sha256sum downloaded-file
gpg --verify downloaded-file.sig downloaded-file
sudo clamscan -r /path/to/test/files
sudo journalctl --since "1 hour ago"
sudo tail -f /var/log/syslog
sudo wireshark
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

5.Fake Wi-Fi Networks at the Republican National Convention, 2016

1. Incident Overview

A security experiment using fake open wireless networks showed that many users would connect to familiar-looking or attractive SSIDs, creating opportunities for traffic interception, identity exposure, and credential theft.

2. Identified Vulnerabilities

- Users trusting SSIDs without verification
- Automatic reconnection to previously seen networks
- Open Wi-Fi and deceptive captive portals
- Evil-twin access points
- Weak certificate validation and user awareness
- Lack of VPN enforcement

3. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

23. Disable automatic connection to open or unknown networks on managed devices.
24. Use WPA2/WPA3 Enterprise for organizational networks and device certificates where practical.
25. Deploy rogue-AP and evil-twin detection at event venues.
26. Use VPN enforcement or secure enterprise connectivity for managed devices on untrusted networks.
27. Train participants to verify network names, captive portals, and certificate warnings.
28. Provide a clearly communicated official SSID and help channel for reporting suspicious networks.

5. Overall Risk Rating

High

6.Commands

```
ip addr
iw dev
iwconfig
nmcli device wifi list
sudo iw dev wlan0 scan
```

```
sudo airmon-ng
sudo airmon-ng start wlan0
sudo airodump-ng wlan0mon
iw dev wlan0mon info
sudo iw dev wlan0mon scan
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.1
sudo nft list ruleset
sudo iptables -L -n -v
sudo ss -tulpn
sudo wireshark
sudo tshark -i wlan0mon -c 100
sudo journalctl --since "1 hour ago"
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

6.Wi-Fi Attacks Against Anti-Doping Officials, 2016

1. Incident Overview

The reported campaign targeting officials using hotel and event Wi-Fi demonstrates the risk of credential theft and unauthorized access to sensitive organizational and medical information during international travel.

2. Identified Vulnerabilities

- Exposure to untrusted hotel/event Wi-Fi
- Credential theft risk
- Insufficient MFA or device authentication
- Weak endpoint protection
- Insecure VPN configuration
- Inadequate travel-security procedures
- Risk to sensitive and medical information

3. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

29. Require phishing-resistant MFA for important accounts where possible.
30. Use device certificates and managed endpoints to strengthen device authentication.
31. Require secure VPN or zero-trust access for sensitive internal applications over public networks.
32. Harden endpoints with EDR, full-disk encryption, patching, application control, and secure configuration.
33. Apply strict travel-security procedures, including avoiding unknown Wi-Fi and reporting suspicious prompts.
34. Encrypt sensitive data and enforce least privilege, retention limits, and access logging.

5. Overall Risk Rating

Critical

6.commands

ip addr
iw dev
iwconfig


```
nmcli device wifi list
sudo iw dev wlan0 scan
sudo iw dev wlan0 scan | grep -E "SSID|BSS|signal|RSN|WPA"
ip route
resolvectl status
cat /etc/resolv.conf
nmcli connection show
nmcli device status
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.1
curl -I http://example.com
curl -I https://example.com
curl -I -L https://example.com
curl -sI https://example.com | grep -i strict-transport-security
openssl s_client -connect example.com:443 -servername example.com
sudo nft list ruleset
sudo iptables -L -n -v
sudo ss -tulpn
sudo tshark -i wlan0 -c 100
sudo wireshark
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

7. Attempted OPCW Wi-Fi Intrusion, 2018

1. Incident Overview

The attempted close-access operation highlighted that enterprise wireless signals can extend beyond physical boundaries and that attackers may position equipment near facilities to target wireless authentication or access.

2. Identified Vulnerabilities

- RF exposure beyond the physical perimeter
- Enterprise Wi-Fi authentication risks
- Improper 802.1X/EAP certificate validation
- Rogue-device and rogue-AP detection gaps
- Insufficient wireless intrusion prevention
- Physical-security weaknesses

3. Vulnerability Assessment

Area	Assessment	Risk
Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

35. Use WPA2-Enterprise or WPA3-Enterprise with 802.1X and strong EAP methods.
36. Validate server certificates correctly on all managed clients to prevent credential exposure to rogue authentication servers.
37. Perform RF surveys to identify signal leakage outside controlled areas.
38. Deploy wireless IDS/IPS to identify rogue APs, suspicious clients, and unusual RF behavior.
39. Integrate wireless alerts with physical surveillance and security response.
40. Use network-access control to verify device identity and security posture before granting access.

5. Overall Risk Rating

Critical

6. commands

```
ip addr
iw dev
iwconfig
nmcli device wifi list
sudo iw dev wlan0 scan
```

```
ip route
resolvectl status
cat /etc/resolv.conf
nmcli connection show
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.10
sudo ss -tulpn
sudo nft list ruleset
sudo iptables -L -n -v
curl -I https://example.com
curl -I -L https://example.com
openssl s_client -connect example.com:443 -servername example.com
sudo journalctl --since "1 hour ago"
sudo ss -tp
sudo tshark -i wlan0 -c 100
sudo wireshark
sudo clamscan -r /home
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

8.Australian Airport and In-Flight Evil-Twin Case, 2024

1. Incident Overview

The reported Australian case involved alleged creation of fake Wi-Fi access points imitating legitimate networks, illustrating how passengers can be tricked into connecting and disclosing personal information.

2. Identified Vulnerabilities

- SSID impersonation and evil-twin access points
- Automatic connection to remembered networks
- Fraudulent captive portals
- Certificate-warning bypass by users
- Passenger data exposure
- Insufficient rogue-AP detection and response

3. Vulnerability Assessment

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Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
Segmentation / Access Control	Verify isolation of guest, corporate, POS, administrative, and sensitive environments using firewalls and least privilege.	High
Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium-High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

41. Use a clearly published official SSID and avoid relying on SSID appearance as the only trust signal.
42. Configure managed devices not to auto-connect to untrusted open networks.
43. Monitor airport/airline wireless space for rogue APs and impersonated SSIDs.
44. Use HTTPS everywhere and ensure captive portals do not request unnecessary sensitive information.
45. Provide passenger guidance explaining how to verify official Wi-Fi and report suspicious portals.
46. Create staff procedures for rapid identification, containment, evidence preservation, and escalation of rogue APs.

5. Overall Risk Rating

High

6.commands

```
ip addr
iw dev
iwconfig
```

```
nmcli device wifi list
sudo iw dev wlan0 scan
sudo iw dev wlan0 scan | grep -E "BSS|SSID|signal|RSN|WPA"
sudo airmon-ng
sudo airmon-ng start wlan0
sudo airodump-ng wlan0mon
ip route
nmcli connection show
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.10
sudo ss -tulpn
sudo nft list ruleset
sudo iptables -L -n -v
sudo tshark -i wlan0mon -c 100
sudo wireshark
sudo journalctl --since "1 hour ago"
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

9. Network Rail Public Wi-Fi Portal Compromise, 2024

1. Incident Overview

The reported incident involved unauthorized content appearing on public Wi-Fi landing pages after misuse of a legitimate administrator account, highlighting the importance of privileged-access security and supplier controls.

2. Identified Vulnerabilities

- Compromise or misuse of a legitimate privileged account
- Insufficient privileged-access management
- Weak or missing MFA
- Third-party supplier access risks
- Insufficient logging and account monitoring
- Weak change approval and configuration control
- Incident-response gaps

3. Vulnerability Assessment

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Wireless / Network Access	Review encryption, authentication, RF exposure, and whether wireless clients can reach sensitive systems.	High
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Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

47. Require MFA for all administrative and supplier accounts, preferably phishing-resistant methods.
48. Use PAM with least privilege, just-in-time access, session recording, and regular entitlement reviews.
49. Separate administrative interfaces from public Wi-Fi user networks.
50. Implement formal change approval, version control, configuration baselines, and rollback procedures.
51. Centralize logs and alert on unusual administrator login locations, times, and configuration changes.
52. Review third-party access contracts, credentials, monitoring, and offboarding procedures.
53. Maintain an incident-response playbook for public-portal compromise and service suspension.

5. Overall Risk Rating

High

6.commands

```
ip addr
iw dev
iwconfig
nmcli device wifi list
sudo iw dev wlan0 scan
sudo iw dev wlan0 scan | grep -E "BSS|SSID|signal|RSN|WPA"
nmcli connection show
nmcli device status
ip route
resolvectl status
cat /etc/resolv.conf
sudo nmap -sn 192.168.1.0/24
sudo nmap -sV 192.168.1.1
curl -I http://example.com
curl -I -L https://example.com
curl -sI https://example.com | grep -i strict-transport-security
openssl s_client -connect example.com:443 -servername example.com
sudo ss -tulpn
sudo nft list ruleset
sudo iptables -L -n -v
sudo tshark -i wlan0 -c 100
sudo wireshark
sudo journalctl --since "1 hour ago"
```

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.

10.Network Rail Public Wi-Fi Portal Compromise, 2024

1. Incident Overview

The reported incident involved unauthorized content appearing on public Wi-Fi landing pages after misuse of a legitimate administrator account, highlighting the importance of privileged-access security and supplier controls.

2. Identified Vulnerabilities

- Compromise or misuse of a legitimate privileged account
- Insufficient privileged-access management
- Weak or missing MFA
- Third-party supplier access risks
- Insufficient logging and account monitoring
- Weak change approval and configuration control
- Incident-response gaps

3. Vulnerability Assessment

Area	Assessment	Risk
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Monitoring	Check centralized logging, rogue-AP detection, IDS/IPS, alerting, and response procedures.	Medium–High
Data / Credential Protection	Assess encryption, session/credential protection, MFA, endpoint controls, and sensitive-data handling.	High

4. Recommended Security Controls

1. Require MFA for all administrative and supplier accounts, preferably phishing-resistant methods.
2. Use PAM with least privilege, just-in-time access, session recording, and regular entitlement reviews.
3. Separate administrative interfaces from public Wi-Fi user networks.
4. Implement formal change approval, version control, configuration baselines, and rollback procedures.
5. Centralize logs and alert on unusual administrator login locations, times, and configuration changes.
6. Review third-party access contracts, credentials, monitoring, and offboarding procedures.
7. Maintain an incident-response playbook for public-portal compromise and service suspension.

5. Overall Risk Rating

High

6.commands

ip addr

iw dev

iwconfig

nmcli device wifi list

ip route

resolvectl status

sudo nmap -sn 192.168.1.0/24

sudo nmap -sV 192.168.1.10

sudo ss -tulpn

sudo nft list ruleset

sudo iptables -L -n -v

curl -I http://example.com

curl -I -L https://example.com

curl -sI https://example.com

openssl s_client -connect example.com:443 -servername example.com

sudo journalctl --since "24 hours ago"

sudo tail -f /var/log/auth.log

sudo tshark -i wlan0 -c 100

sudo wireshark

7.Conclusion

The assessment shows that wireless security cannot be treated as an isolated access-layer issue. Strong encryption and authentication must be combined with segmentation, endpoint protection, monitoring, secure application protocols, privileged-access controls, and an effective incident-response process.